

Quiz Questions:

Quiz:

1. Indentation is crucial in Python to:

- A. Improve code readability
- B. Indicate the end of the program
- C. Separate code blocks
- D. Define the scope of a code block

2. What will be the output of the following code?

```
python
x = 10
if x > 5:
    print("Greater than 5")
else:
    print("5 or less")
```

- A. Greater than 5
- B. 5 or less
- C. Both A and B
- D. None of the above

3. In the `if-elif-else` statement, how many conditions can be checked?

- A. Only one
- B. Two
- C. Multiple
- D. None

4. What is the purpose of the `else` statement in Python?

- A. To handle errors
- B. To provide an alternative block of code when the `if` condition is false

- C. To terminate the program
- D. To declare variables

5. Which of the following statements is true about a nested `if` statement?

- A. It is not allowed in Python
- B. It allows for more complex conditional logic
- C. It always leads to syntax errors
- D. It can only contain one level of nesting

Exercises:

1. **Vowel Checker:**

Write a Python program that takes a character as input and checks whether it is a vowel or not. Use the

`if-else` statement.

2. **Age Group Classification**

Write a program that takes an age as input and classifies the person into one of the following age groups:

- Child: 0-12 years
- Teenager: 13-17 years
- Adult: 18-64 years
- Senior: 65 years and older

3. **Number Classifier:**

Write a program that takes an integer as input and classifies it as positive, negative, or zero. Use the

`if-elif-else` statement.

4. **Leap Year Checker:**

Create a program that checks whether a given year is a leap year or not. A leap year is divisible by 4, but not by 100 unless it is divisible by 400.

5. **Calculator:**

Build a simple calculator program that takes two numbers and an operator (+, -, *, /) as input and performs the corresponding operation.

6. **Short Hand If:**

Rewrite the following code using the short-hand

`if` statement:

```
x = 8
if x % 2 == 0: result = "Even"
else: result = "Odd"
```

7. Discount Calculator:

Create a program that calculates the final price after applying a discount. The program should take the original price and the discount percentage as input.

8. BMI Calculator:

Write a program that calculates the Body Mass Index (BMI) using the formula: $BMI = \text{weight (kg)} / (\text{height (m)})^2$. The program should take weight and height as input.